

PaleoRAG: Frontend Web Application Requirements

Architecture-Grounded UI/UX Specification for the Phylogenetic Context Engine

APPLICATION	TARGET DOMAIN	DOCUMENT VERSION	BACKEND STACK
PaleoRAG	Paleogenomics & Evolutionary Biology	1.0 (Comprehensive)	FastAPI • Qdrant • Ollama • Celery

Purpose of this Document: This requirements specification is prepared directly from the `paleo_rag` repository source code, API contracts, Celery worker task chains, Qdrant vector schemas, evaluation harnesses, and architectural decision records. It serves as an authoritative, self-contained handoff brief for a website-design and development agent to build a production frontend template without needing prior repository inspection.

1. Project Overview

PaleoRAG is a domain-specialized, citation-grounded Retrieval-Augmented Generation (RAG) system tailored for open-access paleogenomics, ancient DNA (aDNA), and evolutionary biology scientific literature. The system addresses critical domain challenges where queries span both vernacular names and extinct taxonomic binomina across deep time horizons.

Core Architectural Pillars

- Hybrid Vector & Lexical Retrieval:** Dense 768-dimensional biological embeddings (`NeuML/pubmedbert-base-embeddings`) coupled with an in-process BM25 lexical index (`rank_bm25`), unified via Reciprocal Rank Fusion (RRF, $k = 60$).
- Dynamic Taxonomy Query Expansion:** Automated query expansion mapping common and fossil taxa (e.g. *saber-toothed cat* → *Smilodon fatalis*; *dire wolf* → *Aenocyon dirus*) via the Global Biodiversity Information Facility (GBIF) Species API and Paleobiology Database (PBDB), cached with 30-day Redis TTL.
- Structure-Aware Document Ingestion:** Parsing PubMed Central Open Access (PMC OA) JATS XML and bioRxiv preprint PDFs, preserving atomic data tables (allele frequencies, loci counts) and figure captions up to 1024 tokens without mid-block slicing.
- Deterministic Deduplication:** SHA-256 content hashing and deterministic UUID5 Qdrant point identifiers for idempotent re-ingestion and stale chunk purges.
- Real-Time Token Streaming & Citation Verification:** Server-Sent Events (SSE) streaming answers with mandatory `[doc_id:chunk_index]` citation markers, followed by a terminal event verifying citations against retrieved chunks to catch hallucinations.
- Verified Benchmark Performance:** Evaluated against a 27-question peer-reviewed paleogenomic golden benchmark (`eval/golden_set.jsonl`) with 100.00% Recall@5 and 100.00% Citation Faithfulness.

2. Feature Inventory

The following table lists every user-relevant feature discovered across the repository backend, its trigger, inputs, outputs, and dependencies.

Feature Name	Description	User Trigger	Inputs Required	Outputs & Dependencies
System Health Monitoring	Verifies live connectivity to Qdrant vector store, Redis broker, and LLM inference engine (Ollama/Anthropic).	Page load, tab focus, or clicking "Refresh".	None (Public endpoint).	Status (<code>ok</code> / <code>degraded</code>), individual service booleans, model tags, and error details. <code>GET /api/health</code> .
Phylogenetic Streaming Chat	Executes taxonomy expansion, hybrid RRF search, LLM answer synthesis, and real-time SSE token delivery.	Submitting a natural language query in the chat bar.	<code>query</code> (str), optional <code>top_k</code> (int, default 8), optional <code>filters</code> (taxon, period, year range).	SSE chunks: <code>{"token": "..."} followed by terminal payload with <code>retrieved_chunks</code> and <code>citation_warnings</code>. <code>POST /api/chat/stream</code>.</code>
Evidence Chunk Inspection	Displays full retrieved passage blocks with section metadata, chunk type, RRF score, and document IDs.	Clicking an inline citation badge or expanding evidence drawer.	Document ID and chunk index (e.g. <code>PMC13453694:0</code>).	Formatted chunk card rendering text, tables (monospace/grid), or figure captions. Source: terminal SSE event.
Citation Hallucination Auditing	Detects and alerts researchers if the LLM cites literature chunks that were not actually retrieved.	Automatic post-processing upon SSE stream termination.	Terminal event <code>citation_warnings</code> array.	Warning banner, flagged citation markers, and explanatory diagnostic text. Computed by <code>check_citations()</code> .
Manual PDF Upload Ingestion	Uploads local scientific paper PDF, parses sections/tables, computes hashes, embeds, and indexes to Qdrant.	Submitting file via drag-and-drop dropzone or file picker.	Binary PDF file (<code>.pdf</code>), <code>source="manual_upload"</code> .	<code>UploadResponse</code> (<code>task_id</code> , <code>status="queued"</code>). Triggers async Celery task chain. <code>POST /api/upload</code> .
PMC / bioRxiv Remote Ingestion	Downloads and indexes open-access papers by PMC accession ID or bioRxiv DOI with strict license verification.	Entering PMC ID or DOI and clicking "Ingest".	<code>doi_or_url</code> (str), <code>source</code> (<code>pmc_oa</code> or <code>biorxiv</code>).	<code>UploadResponse</code> (<code>task_id</code> , <code>status="queued"</code>). Enqueues Celery pipeline. <code>POST /api/ingest</code> .

Feature Name	Description	User Trigger	Inputs Required	Outputs & Dependencies
Ingestion Task Tracking	Polls Celery background execution status across parsing, chunking, dedup, embedding, and vector upsert.	Automatic polling after enqueueing ingestion.	<code>task_id</code> (str).	<code>TaskStatusResponse</code> (<code>state</code> : <code>PENDING</code> , <code>STARTED</code> , <code>SUCCESS</code> , <code>FAILURE</code> ; <code>error</code> ; <code>result</code>). <code>GET /api/task/{id}</code> .
Corpus Manifest Explorer	Browses all indexed scientific papers, license classifications, publication dates, and taxa.	Navigating to "Corpus" view.	None (reads corpus manifest).	List of documents with title, DOI, license badge, publication year, taxon, and status. <i>Requires backend endpoint.</i>
Benchmark & Eval Viewer	Displays verified evaluation metrics (Recall@5, Faithfulness) and model comparison data.	Navigating to "Evaluation" view.	None.	Metrics: 100% Recall@5, 100% Faithfulness, 0 hallucinations, 27 golden questions. <i>Requires backend endpoint.</i>
Bearer Token Auth Config	Manages the secret API bearer token required to authenticate against protected endpoints.	Opening Settings or responding to 401 prompt.	Bearer token string (matches <code>API_BEARER_TOKEN</code>).	Persisted in client storage; injected into <code>Authorization: Bearer <token></code> headers.

3. Required Web Pages / Views

3.1 Primary Research Studio (Chat & Retrieval View)

- **Purpose:** The central user workspace for conducting paleogenomic queries, observing real-time synthesis, inspecting evidence, and validating citations.
- **Key Layout Regions:**
 - *Top Filter Bar:* Taxon selector, Geological Period dropdown, Publication Year range, and Top-K slider.
 - *Conversational Stream:* User questions, assistant markdown responses, live streaming cursor, and interactive citation badges.
 - *Evidence Inspector Drawer:* Collapsible side panel listing retrieved chunks with section context, scores, and chunk type indicators (`text` , `table` , `caption`).
 - *Citation Audit Banner:* High-priority alert banner appearing if any citation warning is emitted.

3.2 Literature Corpus & Ingestion Manager

- **Purpose:** Allows researchers to audit the literature corpus, verify open-access licensing compliance, and ingest new scientific papers.
- **Key Layout Regions:**
 - *Dual Ingestion Hub:* Tabbed panel for (1) Direct PDF file drag-and-drop, and (2) Remote PMC ID / bioRxiv DOI lookup.
 - *Active Ingestion Queue:* Real-time card stack showing active tasks with a 6-stage pipeline progress stepper.
 - *Corpus Catalogue Table:* Filterable data grid of all 32+ ingested studies, showing titles, DOIs, open-access licenses, and indexing status.

3.3 Evaluation & Benchmark Dashboard

- **Purpose:** Provides empirical justification of retrieval quality and citation faithfulness for technical audits.
- **Key Layout Regions:**
 - *KPI Metric Cards:* Recall@5 (100.00%), Citation Faithfulness Rate (100.00%), Hallucinated Citations (0), Active Embedding Model (`PubMedBERT`).
 - *Model Comparison Matrix:* Table comparing Hybrid (PubMedBERT + BM25 + RRF) against Sparse BM25-only baseline.
 - *Golden Benchmark Question Browser:* Interactive table of the 27 golden questions, showing expected source documents and hit status.

3.4 System Health Diagnostics & Settings

- **Purpose:** Infrastructure health monitoring and API credential management.
- **Key Layout Regions:**
 - *Service Connectivity Grid:* Real-time status cards for Qdrant Vector DB, Redis Broker, and LLM Engine (Ollama/Anthropic).
 - *Diagnostics Log & Telemetry:* Detailed error strings and active Ollama model listings.
 - *API Credentials Manager:* Input field for setting and testing the `API_BEARER_TOKEN` .

4. Required UI Elements

Component Scope Rule: Every UI element listed below is strictly justified by the capabilities and data models present in the PaleoRAG repository.

4.1 Navigation & Header Elements

- **App Header:** Displays logo, system name ("PaleoRAG: Phylogenetic Context Engine"), global system health pill (`● Operational` / `▲ Degraded`), active LLM badge, and settings button.
- **View Navigation Tabs:** Seamless tab switching between Research Studio, Literature Corpus, Benchmark Eval, and Diagnostics.

4.2 Inputs & Form Controls

- **Natural Language Query Area:** Multi-line auto-expanding textarea supporting `Enter` to send and `Shift+Enter` for newlines.
- **Taxon Selector (Combobox):** Text input with auto-suggest for scientific binomina (e.g. *Smilodon fatalis*, *Mammuthus primigenius*, *Aenocyon dirus*).
- **Geological Period Dropdown:** Select element containing epoch options (Pleistocene, Holocene, Pliocene, Upper Paleolithic, etc.).
- **Publication Year Range Controls:** Numeric inputs or dual-thumb slider (e.g., 2010 to 2026).
- **Top-K Slider:** Stepped slider control (range: 3 to 20; default: 8).
- **File Dropzone:** Drag-and-drop target accepting `.pdf` files with file validation feedback.
- **Identifier Ingestion Input:** Text field accepting PMC IDs (e.g. `PMC13453694`) or DOIs (e.g. `10.1101/2024.01.02.573861`).

4.3 Specialized Scientific Display Elements

- **Interactive Citation Badge:** Rendered in markdown text as `[doc_id:chunk_index]` (e.g. `[PMC13453694:0]`). Clicking scrolls to and highlights the source chunk card in the evidence drawer.
- **Evidence Chunk Card:**
 - *Header:* Document ID, Chunk Index, and RRF Rank/Score pill.
 - *Section Tag:* Abstract, Introduction, Methods, Results, Discussion, Table, or Caption.
 - *Chunk Type Tag:* `TEXT` (slate), `TABLE` (blue), or `CAPTION` (purple).
 - *Content Body:* Formatted passage text. If `TABLE` , styled in monospace tabular grid. If truncated, shows `[TRUNCATED]` warning marker.
- **Citation Warning Card:** Highlighted alert box identifying hallucinated citation markers (e.g. `[PMC99999:0]`) and diagnostic reason.
- **Ingestion Pipeline Stepper:** Multi-step horizontal progress bar showing: `Parse Document` → `Chunk Document` → `Dedup Check` → `Embed Chunks` → `Upsert to Qdrant` → `Manifest Updated` .

5. End-to-End User Flows

5.1 Primary Flow: Querying, Streaming, and Inline Citation Audit

```
[User enters query: "What spinal pathology was identified in a Smilodon fatalis specimen?"]
    ↓
[UI checks /api/health → confirms Qdrant & LLM are online]
    ↓
[UI issues POST /api/chat/stream with query, filters, and Bearer token]
    ↓
[Backend expands query via GBIF/PBDB → Hybrid Search (PubMedBERT + BM25 via RRF)]
    ↓
[FastAPI streams Server-Sent Events (SSE)]
    • Chunks 1..N: data: {"token": "..."} → UI renders markdown incrementally
    • Terminal: data: {"done": true, "retrieved_chunks": [...], "citation_warnings": [...]}
    ↓
[UI replaces streaming cursor & checks citation_warnings]
    • If empty → displays green "Verified Citation Grounding" badge
    • If non-empty → renders Citation Warning Alert & highlights flagged tags in red
    ↓
[UI populates Evidence Inspector Sidebar with all retrieved chunks]
[User clicks inline citation [PMC13453694:0] → UI highlights Chunk 0 in Sidebar]
```

5.2 Secondary Flow: Literature Ingestion & Progress Tracking

```
[User enters "PMC13453694" into Ingestion Hub and clicks "Ingest Paper"]
    ↓
[UI issues POST /api/ingest → Backend validates license against open-access whitelist]
    ↓
[Backend enqueues Celery chain & returns {"task_id": "task_123", "status": "queued"}]
    ↓
[UI adds task to Active Queue & initiates polling GET /api/task/task_123 every 2.5s]
    • State "PENDING" → Stepper displays "Queued"
    • State "STARTED" → Stepper displays "Parsing & Embedding..."
    • State "SUCCESS" → Stepper displays "Ingested (34 chunks indexed)"
    ↓
[UI automatically refreshes Literature Corpus Table]
```

6. Data & Backend Integration Requirements

6.1 Existing Backend API Endpoints

GET PUBLIC /api/health

- **Description:** Verifies live connectivity to Qdrant, Redis, and Ollama.
- **Response Payload:**

```
{
  "status": "ok",          // "ok" | "degraded"
  "qdrant_ok": true,
  "redis_ok": true,
  "llm_ok": true,
  "detail": {
    "llm_provider": "ollama",
    "ollama_models": ["llama3.2:1b:latest"],
    "qdrant_error": null,
    "redis_error": null,
    "ollama_error": null
  }
}
```

POST AUTH REQUIRED /api/chat/stream

- **Description:** Queries the literature corpus with dynamic taxonomy expansion, hybrid retrieval, and LLM answer streaming.
- **Headers:** Authorization: Bearer <API_BEARER_TOKEN>, Content-Type: application/json
- **Request Payload (ChatRequest):**

```
{
  "query": "What spinal pathology was identified in a Smilodon fatalis specimen from Rancho La Brea?",
  "filters": {
    "taxon_scientific_name": "Smilodon fatalis",
    "geological_period": "Pleistocene",
    "publication_year_min": 2010,
    "publication_year_max": 2026
  },
  "top_k": 5
}
```

- **Response Stream (SSE):**
 - Token events: data: {"token": "A"} • data: {"token": " lumbar"} ...
 - Terminal event:

```
data: {
  "done": true,
  "retrieved_chunks": [
    {
      "doc_id": "PMC13453694",
      "chunk_index": 0,
      "section": "Abstract",
      "chunk_type": "text",
      "score": 0.0327,
      "text": "A lumbar vertebra of Smilodon fatalis from Rancho La Brea exhibited foraminal widening..."
    }
  ],
  "citation_warnings": [
    {
      "cited_marker": "[docZ:9]",
      "reason": "Citation does not correspond to any chunk retrieved for this query (possible hallucination)"
    }
  ]
}
```

POST AUTH REQUIRED /api/upload

- **Description:** Uploads and indexes a local PDF file.
- **Format:** multipart/form-data with field file: UploadFile and source: IngestSource .
- **Response:** {"task_id": "string", "status": "queued"}

POST AUTH REQUIRED /api/ingest

- **Description:** Fetches and indexes a remote paper from PMC OA or bioRxiv.
- **Request Payload (UploadRequest):**

```
{
  "doi_or_url": "PMC13453694",
  "source": "pmc_oa"           // "pmc_oa" | "biorxiv" | "manual_upload"
}
```

- **Response:** {"task_id": "string", "status": "queued"}

GET AUTH REQUIRED /api/task/{task_id}

- **Description:** Returns Celery background task status.
- **Response:**

```
{
  "task_id": "ingest_a1b2c3d4e5f6",
  "state": "SUCCESS",           // "PENDING" | "STARTED" | "SUCCESS" | "FAILURE"
  "error": null,
  "result": { "doc_id": "PMC13453694", "upserted": 34 }
}
```

6.2 Critical Backend Realities & Integration Notes

1. Missing CORS Middleware in `backend/main.py`: The FastAPI application does NOT have CORS middleware enabled. Direct browser requests from a frontend running on another port (e.g. Vite on `:5173` or Next.js on `:3000`) will be blocked by the browser. *Solution:* The frontend template must configure a local reverse proxy (such as Next.js rewrites or Vite proxying) or enable CORS on the backend.

2. Missing Document & Evaluation API Endpoints:

- The corpus manifest exists on disk at `data/processed/manifest.jsonl`, but no `GET /api/documents` endpoint is currently wired in `backend/api/routes.py`.
- Evaluation reports exist at `eval/latest_report.md` and `eval/golden_set.jsonl`, but no HTTP route currently serves them.
- Frontend Design Approach:* The frontend template should define mock fallback services for these views so they can be designed and tested immediately, with clear interfaces to bind to backend routes once added.

3. Authentication Model: Authentication is handled via a single static secret token (`API_BEARER_TOKEN`). There are no user accounts or registration endpoints. The frontend needs an in-app settings modal or environment variable to store and supply this token.

7. Visualization Requirements

- Structured Scientific Tables:** Chunks marked with `chunk_type: "table"` must be formatted in a distinct monospace container or converted into an interactive HTML table with row striping and horizontal scrolling, preserving biological locus data.
- Figure Captions:** Chunks with `chunk_type: "caption"` must be styled with subtle quotation borders, a purple caption badge, and an external link to the publisher's figure repository where available.
- Taxonomy Mapping Network:** Visual display of query expansion: `User Term: "saber-toothed cat"` → `Scientific Binomial: "Smilodon fatalis"` → `Epoch: "Pleistocene"`.
- Benchmark KPI Radial Gauges:** Visual gauges presenting **100.00% Recall@5** and **100.00% Citation Faithfulness** from the official benchmark report.

8. Frontend State Requirements

State	Trigger Condition	UI Representation & Required Behavior
Initial / Empty	Application loaded; no search entered.	Displays clean search interface with domain prompt chips (e.g. "What spinal pathology was identified in Smilodon fatalis?"). Default filters set.
Streaming Active	SSE token stream in progress.	Incremental markdown rendering with a blinking cursor. Submit button toggles to "Stop Generation". Evidence drawer shows loading skeleton.

State	Trigger Condition	UI Representation & Required Behavior
Terminal Success	SSE terminal event with <code>done: true</code> .	Renders full answer, converts citations to interactive tags, populates Evidence Drawer, and displays green "Grounding Verified" indicator.
Citation Warning	Terminal event has non-empty <code>citation_warnings</code> .	Displays amber warning banner: <i>"Citations Flagged: Assistant cited chunks not present in retrieved context."</i> Flagged citation tags styled in red.
No Results / Fallback	0 chunks retrieved or LLM fallback emitted.	Renders assistant message: <i>"The provided literature does not address this."</i> Suggests broadening filters or ingesting additional papers.
Ingestion Polling	Celery task running (<code>PENDING</code> / <code>STARTED</code>).	Active task card displays a 6-stage pipeline stepper indicating the current processing phase.
System Degraded	<code>/api/health</code> returns <code>degraded</code> or unreachable.	Header displays warning pill. If Qdrant is locked or Ollama is offline, displays actionable resolution hints from documentation.
401 Unauthorized	Protected endpoint rejects Bearer token.	Displays modal dialog prompting for the correct <code>API_BEARER_TOKEN</code> .

9. Responsive & Accessibility Requirements

- **Desktop ($\geq 1280\text{px}$):** Three-column view (Left navigation, central chat timeline, right-side persistent Evidence Drawer).
- **Tablet (768px – 1279px):** Collapsible navigation; Evidence Drawer converts to an off-canvas drawer triggered by clicking citation badges.
- **Mobile ($< 768\text{px}$):** Single-column stacked layout; filter bar collapses into a bottom-sheet; tapping citation tags opens a modal bottom-sheet with source text.
- **Accessibility (WCAG AA):**
 - Keyboard traversal through all citation tags and search filters.
 - Inline citations tagged with `aria-label="Citation: Document PMC13453694, chunk 0"`.
 - Stream updates announced via `aria-live="polite"`; citation warnings announced via `role="alert"`.
 - Monospace tables and data chips maintain a contrast ratio $\geq 4.5:1$.

10. Frontend Template Brief (For Design & Development Agent)

The following self-contained brief can be provided directly to a frontend web development agent:

```
APPLICATION: PaleoRAG Web Application
PURPOSE: Citation-grounded search & phylogenetic context engine for paleogenomics & evolutionary biology.
TARGET USERS: Paleogeneticists, evolutionary biologists, and bioinformaticians.

REQUIRED PAGES:
1. Research Studio (/):
  - Natural language search bar with biological filters (Taxon, Geological Epoch, Year range, Top-K).
  - Conversational stream rendering markdown and interactive citation tags ([doc_id:chunk_index]).
  - Evidence Drawer displaying retrieved passage cards (Section, Score, Text/Table/Caption badge).
  - Citation Audit Alert banner alerting user to hallucinated citations.
2. Corpus & Ingestion Manager (/corpus):
  - Ingestion form supporting local PDF drag-and-drop and remote PMC ID / bioRxiv DOI lookup.
  - Active task tracker with 6-stage pipeline stepper (Parse->Chunk->Dedup->Embed->Upsert->Manifest).
  - Filterable data table of ingested scientific papers with open-access license badges.
3. Evaluation Dashboard (/eval):
  - Metric summary cards: Recall@5 (100%), Faithfulness Rate (100%), Hallucinated Citations (0).
  - Model comparison table (PubMedBERT + BM25 RRF vs. Baseline).
  - Interactive 27-question golden set viewer.
4. System Health & Settings (/settings):
  - Real-time status cards for Qdrant Vector DB, Redis, and Ollama LLM.
  - Bearer token authentication manager.

KEY API CONTRACTS & CONSTRAINTS:
- Chat Streaming: POST /api/chat/stream returns Server-Sent Events (SSE). Chunks: {"token": "...".}
  Terminal event: {"done": true, "retrieved_chunks": [...], "citation_warnings": [...]}.
- Ingestion: POST /api/upload (multipart) and POST /api/ingest (JSON) return {"task_id": "...", "status": "queued"}
  Client must poll GET /api/task/{task_id} until state is "SUCCESS" or "FAILURE".
- Authentication: Header "Authorization: Bearer <API_BEARER_TOKEN>".
- Note on CORS: Backend currently lacks CORS middleware; frontend must configure a local dev proxy.
```